Detecting Metaphor in Text

COSI 114a: Fundamentals of NLP

Anna Brezgis

1 Introduction

Metaphor is one of the most fascinating phenomena in natural language. A sentence like "the foundation of our government is strong" makes an underappreciated but compelling rhetorical move: discussing abstract concepts in terms of concrete ones. From a computational perspective, this raises an interesting challenge: can we model these relations without flattening the inference?

In this project, I built a binary classifier to distinguish metaphorical from literal sentences using the LCC dataset. I experimented with traditional bag-of-words features, BERT embeddings, and a hybrid approach combining the best of both.

2 Data

I used the Language Computer Corporation (LCC) Metaphor Dataset (Mohler et al., 2016), specifically the small English subset containing English sentences annotated for metaphoricity. The data was sourced from the ClueWeb09 corpus and the Debate Politics online forum. Each sentence in the dataset has a score indicating how metaphorical annotators judged it to be. Scores of -1 indicate invalid or unclear cases. For valid instances, higher scores indicate stronger metaphoricity.

I filtered out all of the instances with a score of -1, leaving 8,724 valid sentences. I then binarized the labels. Sentences with score ≥ 2 were labeled as metaphorical (1) and sentences with score < 2 were labeled as literal (0).

I split the data 80/10/10 into training, development, and test sets using stratified sampling to preserve the class distribution. The class imbalance (roughly 2:1 literal to metaphorical) is notable and affects model performance, as discussed later.

2.1 Examples from the data

Literal examples:

- "Thank Lyndon Johnson, his Great Society, and the War on Poverty."
- "Taxes on the rich are going up..."

Metaphorical examples:

- "Real enjoyment is at the heart of faith."
- "Kennedy cut taxes..."

Split	Literal (%)	Metaphorical (%)	Total
Train	4,536 (65.0%)	2,443 (35.0%)	6,979
Dev	567 (65.0%)	305 (35.0%)	872
Test	564 (64.6%)	309 (35.4%)	873
Total	**5,667 (65.0%)**	**3,057 (35.0%)**	**8,724**

Table 1: Data distribution across training, development, and test sets after filtering and stratified sampling.

Looking at the data, it becomes clear that one source was a political debate website. A noticable amount of the data references some kind of political or governmental language.

These examples illustrate the challenge: metaphorical sentences are not obviously different in structure or vocabulary from literal ones. The distinction often depends on subtle semantic relationships between words.

3 Dev Set Results

I experimented with three main approaches: bag-of-words (BoW) features with logistic regression as a baseline, BERT embeddings with logistic regression, and a hybrid approach combining both.

1. **Pure BoW:** Unigrams with a frequency of 2+, top 1000 bigrams, and word and sentence length features
2. **BERT:** CLS token embeddings from `bert-base-uncased`, used as a frozen feature extractor

3. **Hybrid:** Concatenation of BoW features and BERT embeddings, with standard scaling applied

For each approach, I tuned the regularization parameter C on the dev set:

Model	C	Dev Accuracy	Dev F1
BoW	0.1	69.95	45.87
BoW	1.0	69.04	**49.63**
BoW	10.0	66.17	47.97
BERT	0.1	66.51	44.27
BERT	1.0	66.74	46.49
BERT	10.0	66.28	**46.55**
Hybrid	0.1	66.86	50.77
Hybrid	1.0	66.97	51.68
Hybrid	10.0	67.78	52.61
Hybrid	100.0	68.35	53.85
Hybrid	500.0	68.69	**54.42**

Table 2: Dev set results for hyperparameter tuning. Best F1 for each model is bolded.

The best dev F1 for BoW is 49.63. There is a slight improvement in the BERT F1 at 46.55, and a marked improvement with the Hybrid model at an F1 of 54.42.

4 Test Set Results

Model	C	Accuracy	F1
BoW	1.0	63.34	46.49
BERT	1.0	65.52	47.10
Hybrid (scaled)	500.0	65.06	**51.20**

Model	C	Precision	Recall
BoW	1.0	48.10	44.98
BERT	1.0	51.54	43.37
Hybrid (scaled)	500.0	50.63	51.78

Table 3: Test set comparison of best configuration from each model.

Class	Precision	Recall	F1
Literal	73.25	72.34	72.79
Metaphor	50.63	51.78	51.20

Table 4: Per-class metrics for best model (Hybrid, C=500).

Metric	Score
Accuracy	65.06
Macro avg F1	62.00
Weighted avg F1	65.15

Table 5: Overall metrics for best model (Hybrid, C=500).

	Pred Literal	Pred Metaphor
True Literal	408 (46.7%)	156 (17.9%)
True Metaphor	149 (17.1%)	160 (18.3%)

Table 6: Confusion matrix for best model (Hybrid, C=500).

5 Discussion

The results reveal interesting patterns about what information matters for metaphor detection. BoW achieved a reasonable performance (dev F1 = 49.63) by capturing lexical associations. For example, common words like "cut" or "break" may appear more often in metaphorical contexts when applied to abstract concepts ("cut ties with them", "break a promise"). BERT embeddings alone, however, performed surprisingly poorly with a dev F1 of 46.55, underperforming the simpler BoW approach. I had expected BERT to excel by an order of magnitude because metaphor understanding seems to require sophisticated semantic knowledge. However, the CLS token captures more generalized sentence-level meaning rather than the specific semantic arithmetic between words that characterizes metaphor. The hybrid approach achieved the best results (dev F1 = 54.42, test F1 = 51.20), suggesting that metaphor detection benefits from both lexical and sentential patterns.

Overall, the results are modest. The test accuracy of 65.06% is only slightly above the 64.6% baseline achieved by always predicting the literal class, and the F1 score of 51.20% indicates that the model catches about half of metaphors while still making substantial errors. The confusion matrix is near-evenly split between error types, with false negatives (metaphors predicted as literal) at 17.1% and false positives (literal sentences predicted as metaphorical) at 17.9%. The near-equal error rate suggests that the model isn't heavily biased towards one class, but also lacks a clear decision boundary between metaphorical and literal language.

6 Conclusion

I built a binary metaphor classifier comparing bag-of-words features, BERT embeddings, and a hybrid approach. The hybrid model achieved the best test F1 of 51.20, modestly outperforming both BoW (46.49) and BERT alone (47.10).

Several aspects proved harder than expected. Most notably, BERT embeddings did not provide the improvement I anticipated. Additionally, the

65/35 class imbalance meant even reasonable accuracy barely beat the majority-class baseline.

With more time, I would like to explore Part-of-Speech tag sequences as an additional feature, or fine-tuning BERT on this specific task rather than using frozen embeddings.

References

[1] Mohler, M., Brunson, M., Rink, B., & Tomlinson, M. (2016). Introducing the LCC Metaphor Datasets. In *Proceedings of the Tenth International Conference on Language Resources and Evaluation (LREC'16)* (pp. 4221–4227). Portorož, Slovenia: European Language Resources Association (ELRA).